

4주차 과제 (c)

사람

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1일차 — 각자 독립 가능한 부분부터

- ☑ C: 메시지 JSON 스키마 설계 (A/B 필요 없음) — 확정되는 대로 A, B한테 공유 ("JS에서 이 필드명으로 `send()` 해줘")

2일차 — 각자 핵심 로직 완성

- ☑ C: C가 만든 스키마 기준으로 메시지 파서 구조 설계 (아직 실데이터 없어 더마로)

3일차 — ★ 인수인계 데드라인 ★

- ☐ C: A의 세션 + B의 후킹 스크립트를 실제로 연결해서 `script.on('message', ...)` 로 첫 실데이터 수신 테스트

▼ `hooks.js` 단독 Test results (성공)

1. `com.google.android.deskclock` (실패)

- frida-server가 32-bit binary인데 deskclock 프로세스는 64-bit로 뜨면서 생긴 아키텍처 불일치
- 2번은 성공해서 이걸 우선 신경 쓰지 않아도 됨.

```
Connected to Android Emulator 5554 (id=emulator-5554)
Spawning `com.google.android.deskclock`
```

```
Failed to attach: unable to handle 64-bit processes due to build configuration
```

2. `com.android.chrome` (성공)

- 명령어: `frida -U -f com.android.chrome -l src/dynamic_analyzer/hooks.js`
- 결과:

```

  ____
 /  _ |  Frida 17.9.1 - A world-class dynamic instrumentation toolkit
| ( _ |
 > _ |   Commands:
/_/  |_  help      -> Displays the help system
      object?    -> Display information about 'object'
      exit/quit  -> Exit
      More info at https://frida.re/docs/home/
      Connected to Android Emulator 5554 (id=emulator-5554)
Spawning `com.android.chrome`...   Spawned `com.android.chrome`. Resuming main th
read!
[Android Emulator 5554::com.android.
chrome ]-> [hooks.js] string_builder / base64 / cipher 후킹 등록 완료
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ad_id': 9570}} data: None
```

```

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E\nnI/tP+MxJTAw9QUWEv9lKtk+jkbqxbsh8nfBUapfKqYn0eidpwq2AzVp3juYl7//fKnaPhJD9gs=\n',
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```



```

IBABt0L074UwLDYKs6Tm8/y\znKkEu116FmH4rkaymUIE0P9KaMftGlmexFlaYjzmB20xZyl6euNXEsQ
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data: None
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ne
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
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'append', 'raw_value': 'Error while loading asset ', 'extra': {}, 'thread_id': 946
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message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:43:44.019Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'assets/stored-locales/en-US.pak', 'extra': {}, 'thread_id':
9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:43:44.020Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'java.io.FileNotFoundException: assets/stored-locales/en-US.
pak', 'extra': {}, 'thread_id': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.213Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'Create MediaDrmBridge with level ', 'extra': {}, 'thread_i
d': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.215Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': ' and origin ', 'extra': {}, 'thread_id': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.215Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': '9BC8071D42F092F2336D4994C9C85DBB', 'extra': {}, 'thread_i
d': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.459Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'Provisioning origin ID ', 'extra': {}, 'thread_id': 9461}}
data: None

```

```

message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.462Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': '9BC8071D42F092F2336D4994C9C85DBB', 'extra': {}, 'thread_i
d': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.468Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'EventListener: No session for event ', 'extra': {}, 'thread
_id': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.517Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'Create MediaDrmBridge with level ', 'extra': {}, 'thread_i
d': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.519Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': ' and origin ', 'extra': {}, 'thread_id': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.522Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': '9BC8071D42F092F2336D4994C9C85DBB', 'extra': {}, 'thread_i
d': 9461}} data: None
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.581Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': 'Security level ', 'extra': {}, 'thread_id': 9461}} data: No
ne
message: {'type': 'send', 'payload': {'hook_type': 'string_builder', 'timestamp':
'2026-07-18T07:44:44.582Z', 'class_name': 'java.lang.StringBuilder', 'method_name':
'append', 'raw_value': ' not supported!', 'extra': {}, 'thread_id': 9461}} data: No
ne

```

3. `com.android.settings` (실패)

- 1번과 같은 이유로 실패함.

▼ `src/dynamic_analyzer/run_test.py` 실행 결과 (실패)

- 명령어: `python -m dynamic_analyzer.run_test` (src 폴더로 이동 후 실행)
- 확인 내용: 파서 쪽 출력이 잘 나오는지. (`[hook_type] method_name → raw_value`)
- 결과: (error)

(1) Error → Frida 최신 버전은 Java를 자동 전역 변수로 취급하지 않음 → require 문법 하나 추가

```
[1] frida-server 연결 시도...
    연결된 디바이스: Android Emulator 5554
[2] com.android.chrome spawn + attach...
    PID: 9706
[3] dynamic_analyzer/hooks.js 로드...
[스크립트 메시지] {'type': 'error', 'description': "ReferenceError: 'Java' is not defined", 'stack': "ReferenceError: 'Java' is not defined\n    at <eval> (/script1.js:117)", 'fileName': '/script1.js', 'lineNumber': 117, 'columnNumber': 1}
[4] 프로세스 재개...
[5] 60초간 관찰 중...
[6] 세션 정리...
완료!
```

(2) Error → Frida-compile Bundling이 아닌 순수 스크립트 파일 하나라서 require 문법 지원 안 됨 → 제거 후 Java 준비될 때까지 재시도 로직 추가

```
[1] frida-server 연결 시도...
    연결된 디바이스: Android Emulator 5554
[2] com.android.chrome spawn + attach...
    PID: 9706
[3] dynamic_analyzer/hooks.js 로드...
[스크립트 메시지] {'type': 'error', 'description': "ReferenceError: 'require' is not defined", 'stack': "ReferenceError: 'require' is not defined\n    at <eval> (/script1.js:6)", 'fileName': '/script1.js', 'lineNumber': 6, 'columnNumber': 1}
[4] 프로세스 재개...
[5] 60초간 관찰 중...
[6] 세션 정리...
완료!
```


▼ 수정한 코드(Line 117)

```
function waitForJava(callback) {
  if (typeof Java !== 'undefined' && Java.available) {
    callback();
  } else {
    setTimeout(function () { waitForJava(callback); }, 50);
  }
}

waitForJava(function () {
  Java.perform(function () {
    hookStringBuilder();
    hookBase64();
    hookCipher();
    console.log("[hooks.js] string_builder / base64 / cipher 후킹 등록 완료");
  });
});
```

(3) 등록 완료 로그 없음 → 재시도 루프가 무한 대기 중일 수 있음(spawn 직후에 스크립트가 주입돼서 계속 실패할 가능성 有)

: A의 `resume()` 호출 시점을 조정하는 게 정공법.

- 스크립트가 로드되지마자 바로 `Java.perform` 을 시도하지 말고 `resume()` 이후에 약간의 지연을 줄 수도 있다고 합니다. 우선 디버그 코드로 확인:

▼ result

[illegible]

```
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[hooks.js] waitForJava 체크: undefined
[6] 세션 정리...
완료!
```

- Java 브릿지가 영원히 주입되지 않음. (구조적인 문제) : CLI와 Python API의 근본적인 차이임. (자동 번들링 유무 차이)
- `frida-compile` 로 미리 번들링

▼ 해결을 위한 환경 설치(cmd, 2개)

1. Node.js 환경 필요(없으면 설치)

```
node --version
```

2. frida-compile 및 frida-java-bridge 설치

```
npm install --save-dev frida-compile  
npm install frida-java-bridge
```

3. hooks.js에서 재시도 로직(`waitForJava`)삭제 후 맨 위에 다시 `require` 추가

```
const Java = require('frida-java-bridge');
```

4. 번들 파일로 컴파일

```
npx frida-compile src/dynamic_analyzer/hooks.js -o src/dynamic_analyzer/hooks.bundle.js
```

5. Python에서 원본 대신 번들 파일 로드

- `run_test.py` 에서

```
JS_PATH = "dynamic_analyzer/hooks.bundle.js" # 원본 대신 번들 파일
```

4일차 — 각자 마무리 + 검증

- ✓ B: 커스텀 XOR 등 표준 API로 못 잡는 패턴 보강, 후킹 노이즈 필터링 보완 (C한테)
- ✓ C: 실제 후킹 데이터 기반으로 필터링(중복 제거, 평문 판별) 로직 완성, `dynamic_report.json` 4차 출력

▼ CLI로 얻은 실행데이터로 만든 로직들

- 명령어: `python -m dynamic_analyzer.message_parser`

```
=== dynamic_report.json 1차 출력 결과 ===
{
  "package_name": "com.android.chrome",
  "session_duration_sec": 0,
  "total_events_captured": 11,
  "total_events_filtered": 10,
  "events": [
    {
      "hook_type": "string_builder",
      "timestamp": "2026-07-18T07:21:56.227Z",
      "class_name": "java.lang.StringBuilder",
      "method_name": "append",
      "raw_value": "18765ebe",
      "extra": {},
      "thread_id": 8930
    },
    {
      "hook_type": "string_builder",
      "timestamp": "2026-07-18T07:21:56.236Z",
      "class_name": "java.lang.StringBuilder",
      "method_name": "append",
      "raw_value": "190e6c42a70",
      "extra": {},
      "thread_id": 8930
    },
    {
      "hook_type": "string_builder",
      "timestamp": "2026-07-18T07:21:56.952Z",
      "class_name": "java.lang.StringBuilder",
      "method_name": "append",
      "raw_value": "Using linker: ",
      "extra": {},
      "thread_id": 8994
    },
    {
      "hook_type": "string_builder",
      "timestamp": "2026-07-18T07:21:56.954Z",
      "class_name": "java.lang.StringBuilder",
      "method_name": "append",
      "raw_value": "org.chromium.base.library_loader.ModernLinker",
      "extra": {},
      "thread_id": 8994
    },
    {
      "hook_type": "string_builder",
      "timestamp": "2026-07-18T07:21:57.541Z",
      "class_name": "java.lang.StringBuilder",
      "method_name": "append",
      "raw_value": "Loaded native library version number \"",
      "extra": {},
      "thread_id": 8994
    }
  ]
}
```

```

{
  "hook_type": "string_builder",
  "timestamp": "2026-07-18T07:21:57.544Z",
  "class_name": "java.lang.StringBuilder",
  "method_name": "append",
  "raw_value": "83.0.4103.106",
  "extra": {},
  "thread_id": 8994
},
{
  "hook_type": "base64",
  "timestamp": "2026-07-18T07:21:57.909Z",
  "class_name": "android.util.Base64",
  "method_name": "encodeToString",
  "raw_value": "MIIEqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSIb3DQEBAUAMIGU...",
  "extra": {
    "direction": "encode"
  },
  "thread_id": 8930
},
{
  "hook_type": "base64",
  "timestamp": "2026-07-18T07:21:57.910Z",
  "class_name": "android.util.Base64",
  "method_name": "decode",
  "raw_value": "MIIEqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSIb3DQEBAUAMIGU...",
  "extra": {
    "direction": "decode"
  },
  "thread_id": 8930
},
{
  "hook_type": "string_builder",
  "timestamp": "2026-07-18T07:21:58.418Z",
  "class_name": "java.lang.StringBuilder",
  "method_name": "append",
  "raw_value": "Error while loading asset ",
  "extra": {},
  "thread_id": 8930
},
{
  "hook_type": "string_builder",
  "timestamp": "2026-07-18T07:21:58.420Z",
  "class_name": "java.lang.StringBuilder",
  "method_name": "append",
  "raw_value": "assets/stored-locales/en-US.pak",
  "extra": {},
  "thread_id": 8930
}
],
"plaintext_candidates": [
  "18765ebe",
  "190e6c42a70",
  "Using linker: ",
  "org.chromium.base.library_loader.ModernLinker",

```



```

    "Loaded native library version number \"",
    "83.0.4103.106",
    "MIIEqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSib3DQEBAUAMIGU...",
    "MIIEqDCCA5CgAwIBAgIJANWFuGx90071MA0GCSqGSib3DQEBAUAMIGU...",
    "Error while loading asset ",
    "assets/stored-locales/en-US.pak"
  ],
  "errors": []
}

```

원본 11개 → 필터링 후 10개
 평문 후보 10개 발견

▼ 수정 필요한 부분(아직 업뎃 안함)

5일차 — 전원 통합

- ☐ 전원이 실제 앱(정상 1개 + 악성 1개)으로 A→B→C→D 전체 파이프라인 end-to-end 실행
- ☐ 버그 수정, `dynamic_report.json` 최종 형식 확인
- ☐ 발표자료(데모 캡처, 실행 로그) 준비